

NexGen High Precision

Tools for Composite Machining



AXIS have the right Milling and Drilling solutions designed for machining difficult to machine CFRP (Carbon-Fiber Reinforced Plastic) and MMC (Metal Matrix Composites). Composites and especially Carbon fibres are being utilized increasingly due to their favourable physical and mechanical properties.

Problems while machining Composites :

- High Cutting forces
- Abrasive Materials
- Machined surface quality
- Dimensional stability

Delamination

Rapid Tool wear

Spalling

Uncut fibers

Uncut Resin

These problems are caused by unique conditions created by composite materials, which make it very difficult to get smooth surfaces.

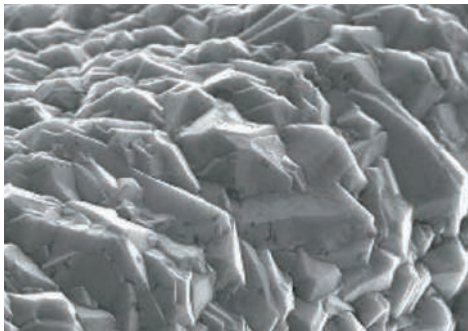
AXIS Tools for CFRP are optimally engineered for these applications to counter the machining challenges.

AXIS Tools for CFRP have special geometries and coatings with following features:

- Specially designed geometries
- Sharp cutting edges with optimised grinding techniques
- NexGen proprietary CVD Diamond coatings



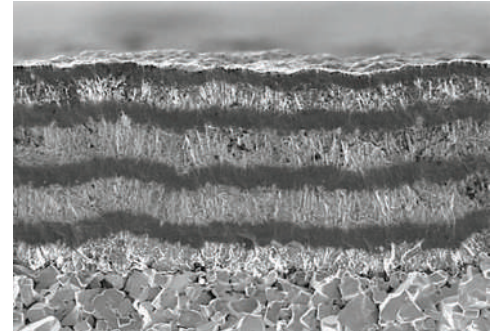
NexGen CVD Diamond Coatings



Microcrystalline Diamond Coating

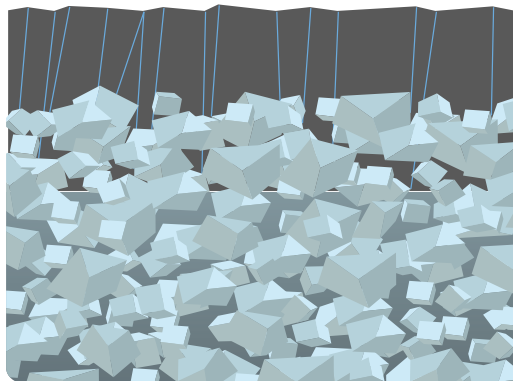


Nano-Crystalline Diamond Coating



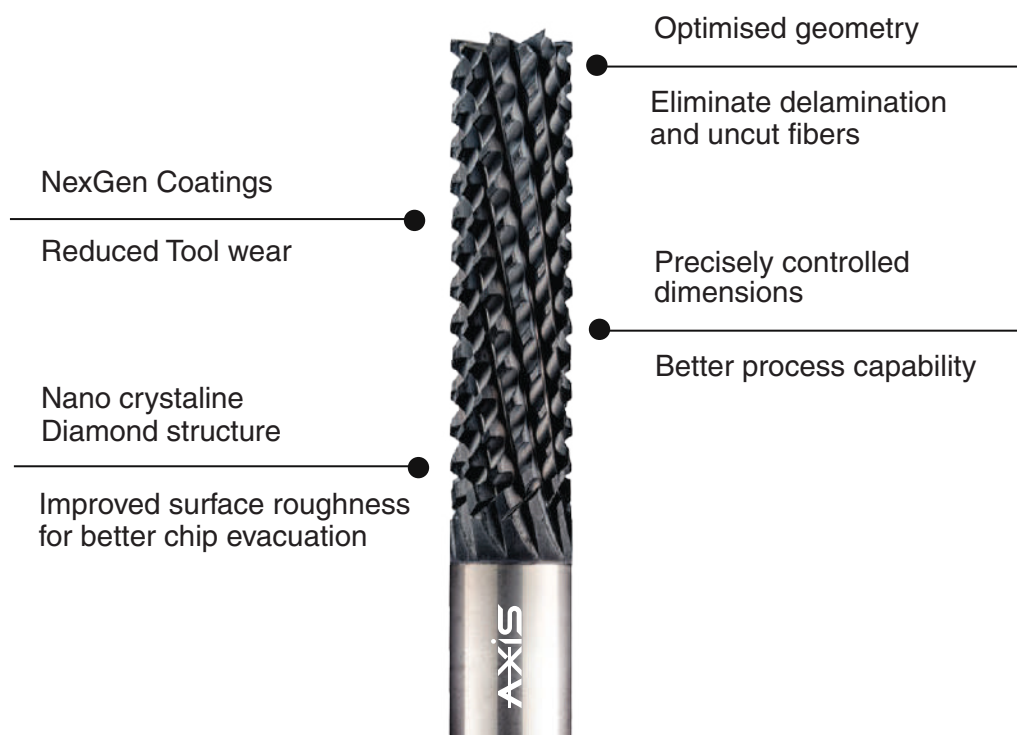
Multilayer Diamond Coating

Excellent adhesion and very smooth surfaces



- Cobalt 
- Tungsten Carbide 
- Diamond 

Multilayer Diamond Coatings have excellent adhesion because they are adapted to the carbide, geometry and application. At the same time, they form very smooth surfaces.



Tools for Composite Machining

Article	Tool Drawing		Series	Dia Range (mm)	Helix
P0661D			CFRP Chip Breaker Router	3.00 - 20.00	St. Helix Right Cut
P0681D			CFRP Chip Breaker Router	3.00 - 20.00	Right Helix Right Cut
P0671D			CFRP Chip Breaker Router	3.00 - 20.00	Left Helix Right Cut
P0901D			CFRP Diamond Cut Router Drill End	2.00 - 20.00	Right Helix Right Cut
P0401D			CFRP Diamond Cut Router Drill End	2.00 - 20.00	Left Helix Right Cut
P0801D			CFRP Diamond Cut Router Safe End	2.00 - 20.00	Right Helix Right Cut
P0501D			CFRP Diamond Cut Router Safe End	2.00 - 20.00	Left Helix Right Cut
P1011D			CFRP Diamond Cut Router Burr End	2.00 - 20.00	Right Helix Right Cut
P1021D			CFRP Diamond Cut Router Burr End	2.00 - 20.00	Left Helix Right Cut
P0701D			CFRP Diamond Cut Router Ball End	2.00 - 20.00	Right Helix Right Cut
P0711D			CFRP Diamond Cut Router Ball End	2.00 - 20.00	Left Helix Right Cut
P0111D			CFRP Diamond Cut Router Flat End	2.00 - 20.00	Right Helix Right Cut
P0121D			CFRP Diamond Cut Router Flat End	2.00 - 20.00	Left Helix Right Cut
B9721D			CFRP 4F Ball Endmill	6.00 - 20.00	Right Helix Right Cut
E1171D			CFRP 4F Endmill with Corner Radius	6.00 - 20.00	Right Helix Right Cut
P1001D			CFRP Compression Router	6.00 - 20.00	Right Cutting

T660			INDrill CFRP	0.50 - 20.00	Right Helix Right Cut
T0311			Calibration Drill	2.00 - 20.00	Straight Helix Right Cut
T4991			Kevlar Drill	2.00 - 20.00	Right Helix Right Cut

Composite Type			Machining Operations						Machining Type	
CFRP	GFRP	Honey Comb	Slotting	Side Milling	Surface Milling	Profiling	Roughing	Finishing	Hand	CNC
●	●	◐	●	●	●	○	●	●	○	●
●	●	◐	●	●	●	○	●	●	○	●
●	●	◐	●	●	●	○	●	●	○	●
●	●	○	●	●	○	○	●	●	●	●
●	●	○	●	●	○	○	●	●	●	●
●	●	○	●	●	●	○	●	●	●	●
●	●	○	●	●	●	○	●	●	●	●
●	●	○	◐	●	◐	○	●	●	●	●
●	●	○	◐	●	●	○	●	●	●	●
●	●	○	●	●	○	●	●	●	●	●
●	●	○	●	●	○	●	●	●	●	●
●	●	○	○	●	○	○	●	●	●	●
●	●	○	○	●	○	○	●	●	●	●
●	●	●	●	●	○	●	◐	●	○	●
●	●	●	●	●	○	◐	◐	●	○	●
●	◐	●	◐	●	◐	○	○	●	○	●

Composite Type				Machining Type	
CFRP	GFRP	Kevlar	Honey Comb	Hand	CNC
●	●	○	○	●	○
●	●	○	●	●	○
○	○	●	○	●	○

Description : ● Optimal ◐ Suitable ○ Not Suitable

Composite Machining



AXIS
Tools for **m**ICRO **m**ACHINING



IND - SPHINX PRECISION LTD (Unit B)
1 Taksal Road Parwanoo - Kasauli Marg
Parwanoo Himachal Pradesh India 173220
☎ +91 1792 234460 / 232860 ☎ +91 1792 232898
✉ info@axis-microtools.com 🌐 www.axis-microtools.com



AXIS Europe GmbH
Danziger Str. 3, 88250 Weingarten
☎ +49 (0)751 560 1589 - 0
✉ info@axis-europe.eu 🌐 www.axis-europe.eu



QUALITY • PRECISION • CONSISTENCY

